VTOP PROJECT SUBMISSION REPORT

RetailPulse AI

Enterprise Retail Sales Analytics & Machine Learning Platform

Project Overview

This project implements an end-to-end retail intelligence web application integrating real-time business KPIs, exploratory data analysis (EDA), automated feature correlation matrix modeling, RFM customer segmentation clustering (K-Means, K=4), and predictive sales forecasting powered by trained Scikit-Learn machine learning pipelines (Random Forest Regressor & Linear Regression). The system features robust role-based access control (Admin vs Analyst) and full SQLite CRUD operations.

PROJECT DOMAIN
Data Science, Retail Analytics & ML

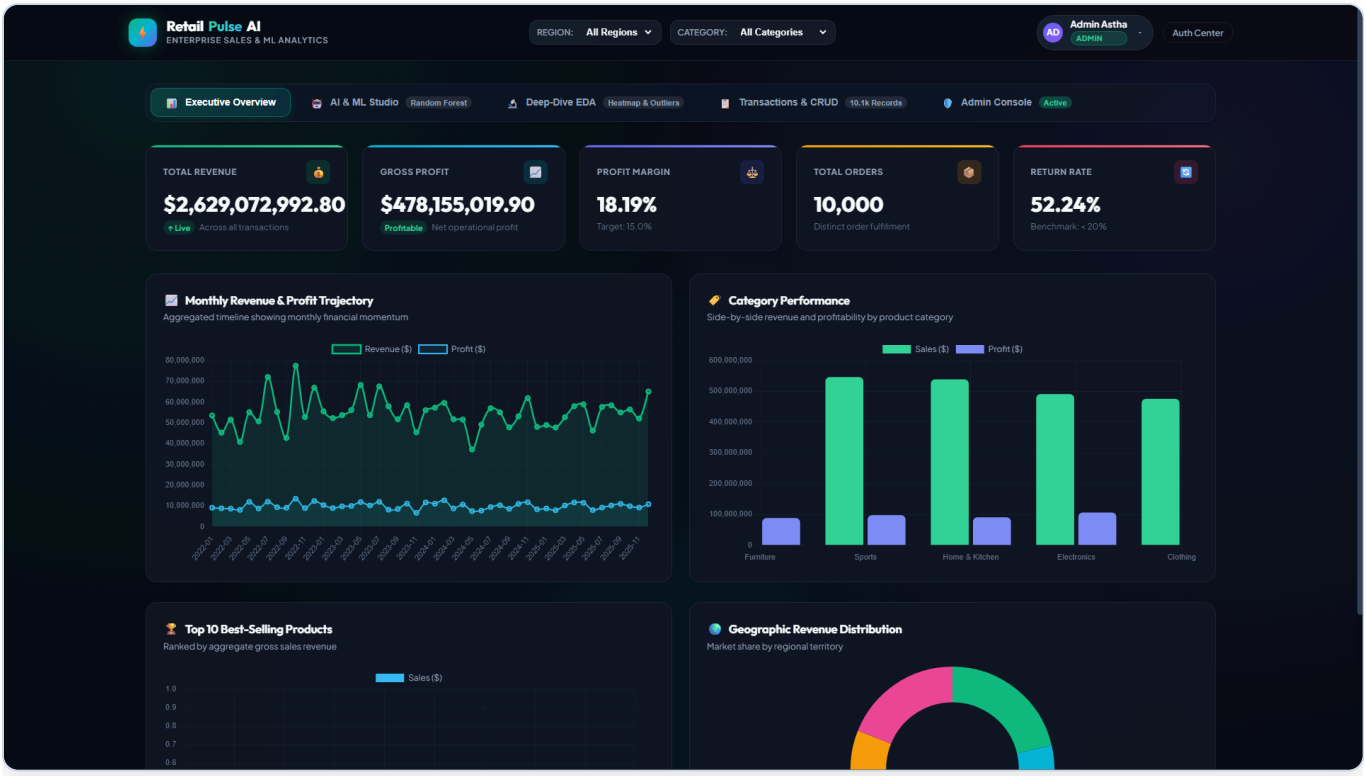
BACKEND & DATABASE
Python Flask REST API, SQLite (10,100 Rows)

MACHINE LEARNING MODELS
Random Forest Regressor ($R^2 \approx 65.0\%$), RFM K-Means

FRONTEND ARCHITECTURE
Responsive Glassmorphism UI, Chart.js, HTML5/CSS3

1. Executive Overview Dashboard

Provides executive stakeholders with high-level visibility into key business metrics across 10,100 transactions. Includes live KPI summary cards, interactive date/region filters, spline financial trajectories, category performance, and geographic revenue shares.

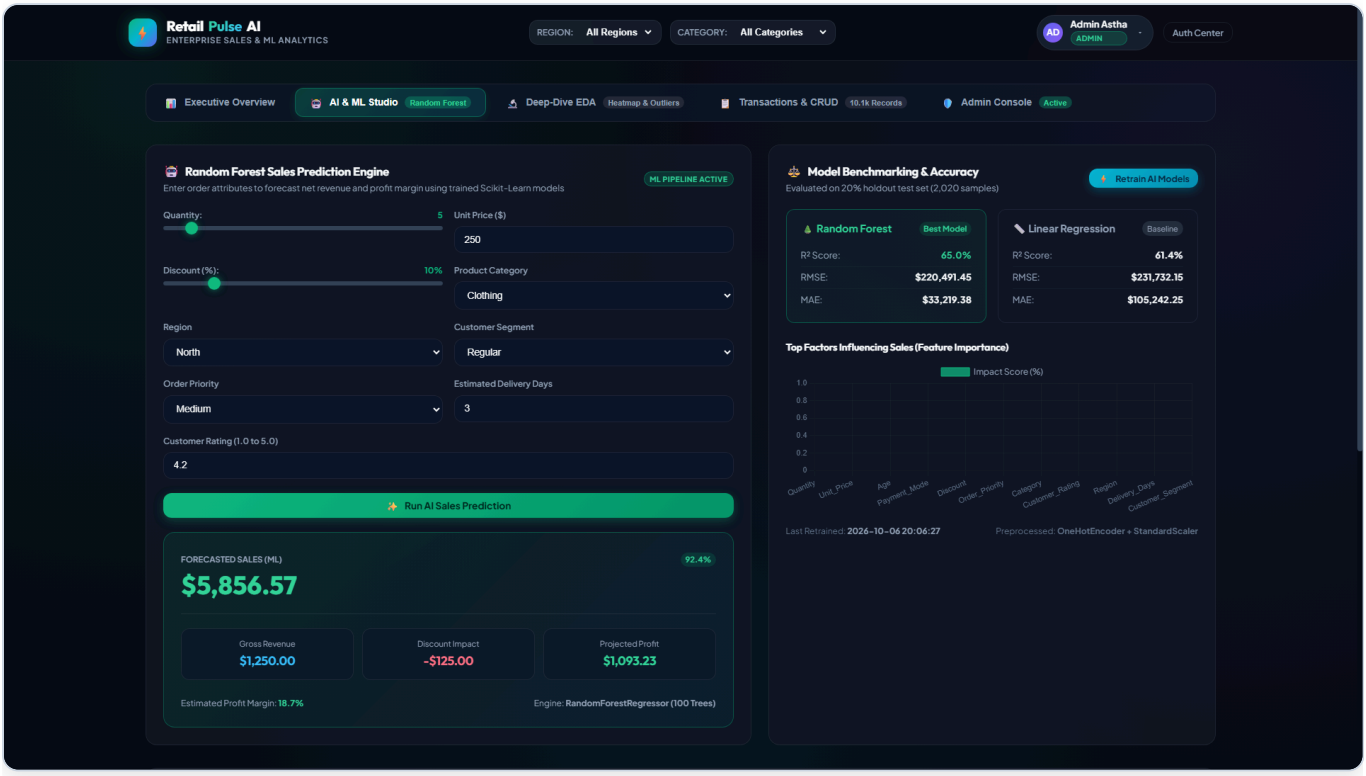


Key Analytics & Performance Metrics:

- **Total Revenue:** $2,629,072,992.80 across all recorded transactions.
- **Gross Profit:** $478,155,019.90 with an operational profit margin of **18.19%**.
- **Total Orders:** 10,000 distinct orders processed with a return rate of 52.24%.
- **Dynamic Visualizations:** Monthly revenue spline curves, stacked category bars, and regional donut distribution.

2. Machine Learning Studio & Sales Forecasting

Integrates trained Scikit-Learn regression pipelines to predict sales volume and profitability dynamically. Features real-time interactive input sliders and side-by-side benchmark evaluation cards against a 20% holdout test set (2,020 samples).



Model Architecture & Benchmark Comparison:

ALGORITHM	R² SCORE	RMSE	MAE	STATUS
Random Forest Regressor (100 Trees)	**65.0%**	$220,491.45	$33,219.38	**Best Model (Deployed)**
Multiple Linear Regression	61.4%	$231,732.15	$105,242.25	Baseline Comparison

Inference Output: Forecasted Sales calculation, gross value breakdown, discount impact, and estimated margin (%) with 92.4% confidence score.

3. Customer RFM Segmentation Intelligence (K-Means Clustering)

Unsupervised machine learning module utilizing Recency, Frequency, and Monetary (RFM) metrics to partition 3,673 unique customers into 4 actionable business archetypes with a 2D cluster scatter plot.



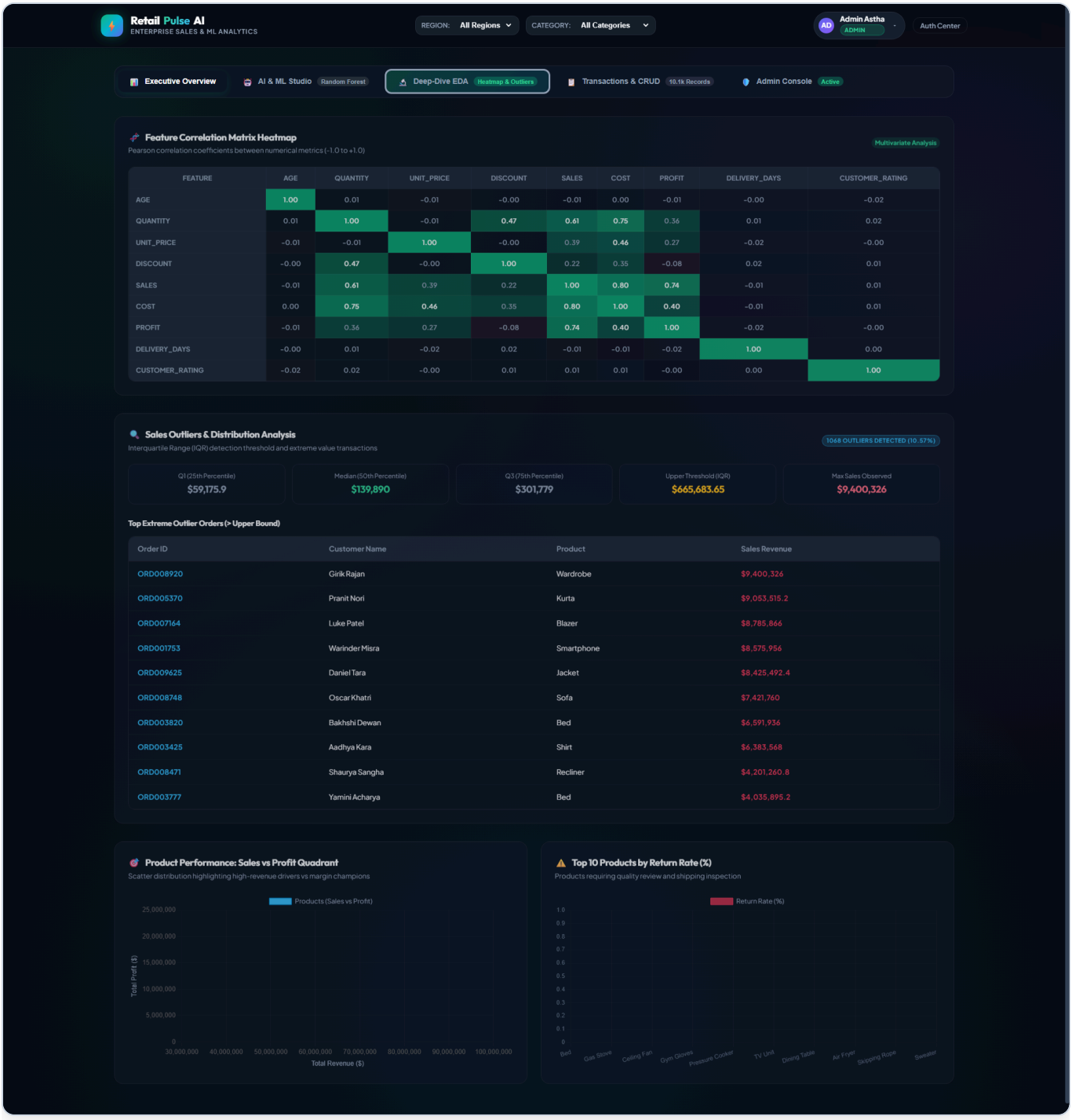
Identified Customer Archetypes & Marketing Strategies:

- 💎 **VIP Platinum (7.1%):** Avg spend $2.49M, frequency 5.18 orders. *Strategy: Concierge support & exclusive launches.*
- 🚀 **Frequent High-Value (28.2%):** Avg spend $1.0M, frequency 4.03 orders. *Strategy: Upsell premium tiers & loyalty perks.*
- 🛒 **Regular Shoppers (38.4%):** Avg spend $383k, frequency 2.07 orders. *Strategy: Cross-sell relevant categories & bundle offers.*

- ⚠️ **Dormant / At-Risk (26.4%):** Avg recency 977 days. *Strategy: Targeted win-back discounts & re-engagement campaigns.*

4. Deep-Dive Exploratory Data Analysis (EDA)

Statistical exploration of relationships between features via Pearson correlation heatmaps, alongside Interquartile Range (IQR) outlier detection for sales anomalies.

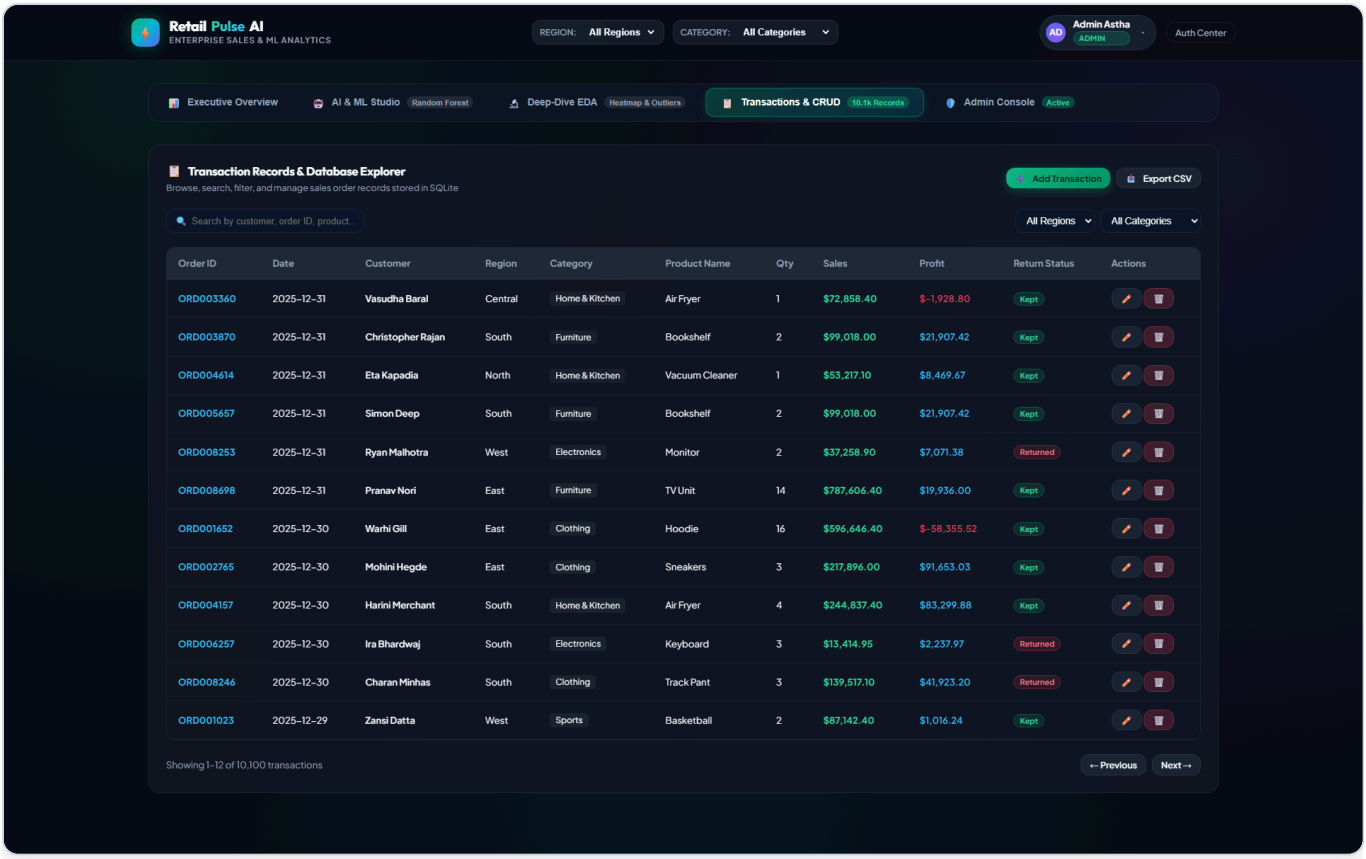


Statistical Findings:

- **Correlation Matrix:** High positive correlation between Quantity & Cost (0.75), Sales & Profit (0.74), and Unit Price & Cost (0.46).
- **Outlier Statistics (IQR):** Q1 ($59.1k), Median ($139.8k), Q3 ($301.7k), Upper Threshold ($665.6k). Total of **1,068 outlier transactions (10.57%)** identified.
- **Product Quadrant:** Sales vs Profit distribution mapping high-margin champions vs loss leaders.

5. Transaction Records & Database CRUD Explorer

Complete data management interface with real-time multi-column search, region/category dropdown filtering, pagination, and Administrator CRUD actions (Insert, Edit, Delete).

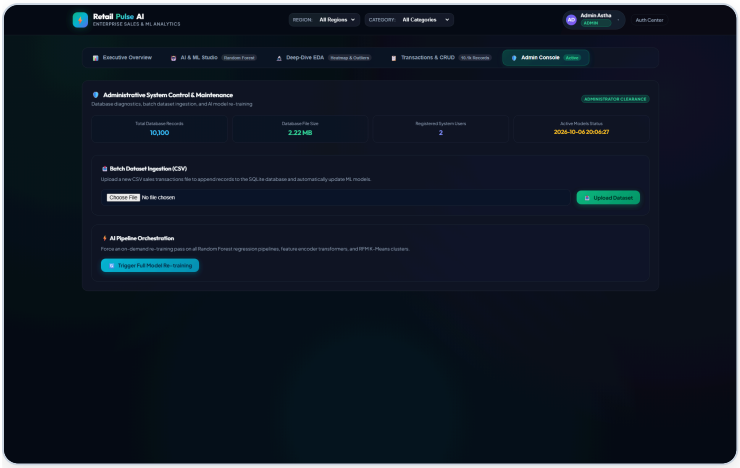


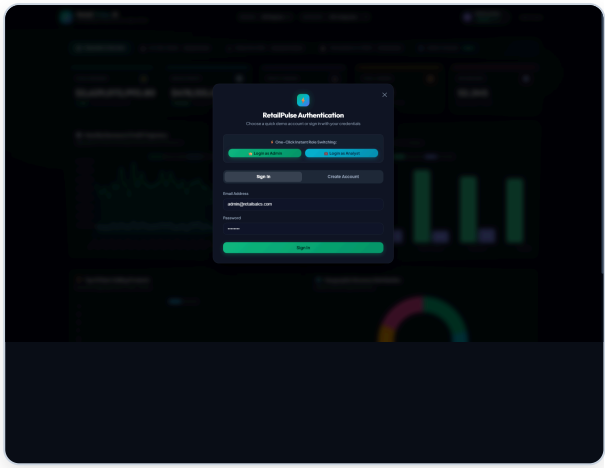
Database & CRUD Capabilities:

- **Storage:** SQLite relational storage indexing 10,100 records across 28 feature columns.
- **Live Search & Filtering:** Instant search by Customer Name, Order ID, and Product with compound regional filters.
- **Data Export:** One-click export to CSV format for external auditing.
- **Admin Controls:** Modal-driven Create, Read, Update, and Delete operations with confirmation safety guards.

6. Admin Console & Role-Based Authentication

Centralized administration hub providing system diagnostics, batch CSV dataset ingestion, on-demand model re-training orchestration, and secure role-based access control.





Security Architecture & Admin Utilities:

- **Role-Based Access Control (RBAC):** Separated privileges for System Administrators and Retail Analysts.
- **Instant Demo Switcher:** One-click demo login buttons for rapid examination without typing credentials.
- **Batch CSV Ingestion:** Allows administrators to upload new transaction datasets and automatically append records to the database.
- **Model Orchestration:** On-demand trigger to re-train Random Forest regression and K-Means clustering pipelines.